

Lean fidelity repair

PRIME.RDAGGER.LEAN_FIDELITY_REPAIR.01 — round 463

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Abstract

Round 462 found that the selected Lean window was a raw-prime-power cosine quotient rather than the production lag/spectral-density fold. Round 463 repairs the object, pins it numerically at $k = 5, 9, 10$, and turns the three absent external arrows into named Lean obligations. The honest `sorry` census rises from five to eight. This is an internal formalization repair, not a claim of the Riemann hypothesis.

1 Productive transcription

`RH/FaithfulFold.lean` implements

$$\{\Lambda(n) : n \leq a^2\} \longrightarrow c_i^P \longrightarrow c_i = c_i^A + c_i^P \longrightarrow d_j \longrightarrow (x_j, w_j) \longrightarrow (\mu, \nu),$$

where

$$M = m + 1, \quad L = 2M - 2, \quad d_j = c_0 + 2 \sum_{r=1}^{M-2} c_r \cos(r\theta_j) + c_{M-1} \cos((M-1)\theta_j),$$

$$\theta_j = \frac{2\pi j}{L}, \quad x_j = \cos \theta_j, \quad w_j = \frac{2}{L}(1 - \cos \theta_j)d_j, \quad 1 \leq j \leq L/2.$$

The endpoint $j = L/2$ is halved. Positive weights form the `combWeight` channel and negative weights, with sign reversed, form `archWeight`; therefore `PrimeWindow.combHankel` is the production μ -Gram matrix. Zero weights are retained in Lean; Python's 10^{-300} deletion threshold is not mathematical data.

The prime lag is an exact finite sum of ordinary and reflected triangular tents with mass $\frac{1}{2}\text{combMass}(n) = \Lambda(n)/\sqrt{n}$. `productionArchLag` exposes each value of `arch_lags(M, Delta)`, and `ProductionCompletion` exposes the `v958` border and `r243` budget. Their analytic construction remains open, but no per-atom `canonicalCompletion` enters the selected window.

2 Numerical seal

Because Lean's `Real.log` and `Real.cos` are noncomputable, the verifier pins exported evaluations; no `native_decide` or sampled proposition is embedded.

k	atoms	S	cap	H_{00}^μ	H_{01}^μ	H_{11}^μ
5	198	19	10	1.03660220661	-0.725897612286	0.604943507078
9	23150	71	36	1.90736517945	-1.15683291482	1.06483617754
10	82267	79	40	1.91210951462	-1.15236092729	1.03066236713

The independent explicit cosine sum agrees with the production FFT, and all displayed values agree with the production builder within $2 \cdot 10^{-12}$. Two smoke runs are byte-identical; wall time is below six seconds.

3 Honest path graph

The internal endpoint is now `internal_weil_nonneg_of_frequently_selected`, whose conclusion is only

$$\forall f : \text{GridElement}, \quad 0 \leq \text{weilForm}(f).$$

The proposition `FrequentlySelectedInternalMincut` contains both the spectral `FREQ` hypothesis and the `AXIOM-CANDIDATE SelectedACapPsdImpliesPlainReads`. Thus the read-identification remainder is no longer easy to omit when quoting the mincut.

The three arrows beyond the internal endpoint are declarations in `RH/ExternalBridges.lean`:

1. `grid_dense_extension`: native dyadic autocorrelations to the full Weil test class;
2. `standard_explicit_formula_identification`: the continued custom form equals the standard explicit formula;
3. `standard_weil_criterion_to_mathlib_rh`: standard positivity implies Mathlib’s actual `RiemannHypothesis`, defined using `riemannZeta`.

Each carries one `sorry`; no theorem named “RH from frequently selected” remains.

4 Sorry census

Declaration	r463 status	Reason
<code>lstar_canonical</code>	unchanged	Alternative rational-certificate route.
<code>terminal_q_canonical</code>	unchanged	Alternative terminal route.
<code>pair_terminal_dictionary</code>	unchanged	Measured dictionary, off internal path.
<code>mainWindow_iff_</code>	unchanged	Historical opaque source bridge.
<code>builtFromPrimeSource</code>		
<code>arch_elementwise_</code>	unchanged	Gauss/Mellin/digamma identification; <code>archRead</code> itself is now concrete.
<code>stabilization</code>		
<code>grid_dense_extension</code>	new	Previously absent density and continuity theorem.
<code>standard_explicit_</code>	new	Previously absent normalization and zero-side dictionary.
<code>formula_identification</code>		
<code>standard_weil_criterion_</code>	new	Previously absent bridge to Mathlib’s zeta/RH predicate.
<code>to_mathlib_rh</code>		

5 Kill and gate table

Gate	Result	Evidence
Production dimensions and sign split	PROVED	Lean definitions; $L = 2M - 2$.
$k = 5, 9, 10$ Lean-mirror versus Python	PROVED	11/11 probe gates; tolerance $2 \cdot 10^{-12}$.
No weakening of	PROVED	Full <code>lakebuild</code> .
Loewner/whitening/FREQ layer		
Concrete arch read	PROVED	Finite lag pairing; coefficient theorem remains open.
Selected A-cap to plain reads	TYPED OPEN	Explicit member of compound mincut.
External RH bridge	TYPED OPEN	Three named <code>sorrys</code> , actual Mathlib interface.
Sorry census	PROVED	Five old plus three new, total eight.

Machine checks. `lean_fidelity_probe.py--smoke`: LEANFIDELITYVERIFIED. `verify_lean_fidelity.py`: LEANFIDELITYREPAIRVERIFIED. `lakebuild`: green.

Claim boundary. Research documentation. No L^* claim, no $R^\dagger$ claim, and no claim for or against the Riemann hypothesis.